

No class on Fridays

We'll use 3 of the 4 days M-TH 2-3.15

Next week M W Th

So far: Vectors in $\mathbb{R}^n$, matrices to solve linear equations

Now: abstract vector spaces (& soon linear maps)

We will need abstract reasoning based on axioms

Today's concepts: span, linear (in)dependence
basis, dimension

Caution: formalism is useful and essential at CMI
but is NOT an end in itself. Do NOT
be seduced by it. (H. Weyl quote in preface)

① V. space: set with addition, subtraction,
scalar mult, satisfying some properties

Examples: $\mathbb{R}^n$, all polynomials, poly of $\deg \leq n$
poly in 2 (or any fixed number of) variables

② Subspace of v. space V = subset closed under $+$, $\cdot$ (inherits
properties of operations from V & becomes a v. space in its
own right)

Examples: homogeneous poly of degree d in k variables
e.g. $\{ax^2 + bxy + cy^2 \mid a, b, c \in \mathbb{R}\} \subset$ all poly in
 x & y

Subspaces associated with $r \times c$ matrix A

i.e. with function $f(x) = Ax$ ($f: \mathbb{R}^c \rightarrow \mathbb{R}^r$)

$$\ker f = \{v \mid Av = 0\} \subset \mathbb{R}^c$$

$$\operatorname{im} f = \{Av \mid v \in \mathbb{R}^c\} \subset \mathbb{R}^r$$

③ Span of a set of vectors $S = \{v_1, \dots, v_n\} \subset$ v.space V

$$= \{c_1 v_1 + \dots + c_n v_n \mid c_i \in \mathbb{R}\} = \text{smallest subspace of } V \text{ containing } S$$

Examples: poly of $\deg \leq n = \text{span of } 1, x, x^2, \dots, x^{n-1}$

For $A = r \times c$ matrix $f(x) = Ax$, range of $f = \text{span of columns of } A$

$$\left(\text{b/c } \begin{bmatrix} | & | & \dots & | \\ v_1 & v_2 & \dots & v_c \\ | & | & \dots & | \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 v_1 + \dots + x_c v_c \right) \text{ Thus } f \text{ is surjective} \\ \Leftrightarrow \text{cols of } A \text{ span } \mathbb{R}^r$$

Note: f is injective $\Leftrightarrow (Ax = 0 \Rightarrow x = 0)$

$$\Leftrightarrow (x_1 v_1 + \dots + x_n v_n = 0) \text{ only when all } x_i = 0$$

④ (multi)set of vectors $v_1, v_2, \dots, v_n$ is called

linearly independent if $(x_1 v_1 + \dots + x_n v_n = 0 \Rightarrow \text{all } x_i = 0)$

linearly dependent if we can find $x_1, x_2, \dots, x_n$ at least one of which is nonzero such that $\sum x_i v_i = 0$

Examples • We saw columns of A indep $\Leftrightarrow f(x) = Ax$ is injective

• Any set of vectors containing 0 is lin dependent (b/c $1 \cdot \vec{0} = \vec{0}$)

- Any nonzero vector is lin. indep.
- Any multiset in which a vector repeats is dependent
- v_1, v_2 is lin dependent $\Leftrightarrow av_1 + bv_2 = 0$ a, b not both 0
 $(\text{wlog } a \neq 0) \quad v_1 = -\frac{b}{a}v_2$
 $\Leftrightarrow$ one of v_1, v_2 is a scalar multiple of the other

Useful observation $v_1, v_2, \dots, v_n$ is dependent $(n > 1)$
 $\Leftrightarrow$ some v_i is a linear combination of others

Proof:

$$\Leftarrow \text{ If } v_i = c_1 v_1 + \dots + c_{i-1} v_{i-1} + c_{i+1} v_{i+1} + \dots + c_n v_n$$

$$\text{Then } \sum c_i v_i = 0 \text{ where } c_i = -1$$

$$\Rightarrow \text{ Given } \sum c_i v_i = 0 \text{ not all } c_i = 0 \text{ wlog } c_1 \neq 0$$

$$\text{Then } v_1 = \sum_{i=2}^n -\frac{c_i}{c_1} v_i$$

Observe: (a) If S spans V , then any set containing S also spans V

(b) If S is indep, then so is any subset of S

- Any spanning set in $\mathbb{R}^n$ (i.e. set whose span is $\mathbb{R}^n$) must be of size at least n .

Pf. If $v_1, v_2, \dots, v_k$ spans $\mathbb{R}^n$, consider $n \times k$ matrix $\begin{bmatrix} | & | & & | \\ v_1 & v_2 & \dots & v_k \\ | & | & & | \end{bmatrix} = A$. Then $x \mapsto Ax$ is surjective $\mathbb{R}^k \rightarrow \mathbb{R}^n$

$\Rightarrow$ Every row of A must have a pivot

$$\Rightarrow n \leq k$$

- Any linearly independent set in $\mathbb{R}^n$ must be of size $\leq n$
Pf. Use the same matrix

⑤ Def: A basis of a vector space V is an independent set that spans V .

"The" cardinality of a basis of V is called

"the" dimension of V . [Need to justify "the" by showing that a basis exists & any two have equal cardinality]

We saw above : Any basis of $\mathbb{R}^n$ must contain exactly n elements $\therefore$ dimension of $\mathbb{R}^n$ is n

Exercise: Given matrix A , how will you find basis of nul space & col space of A

Do this!